



DEPARTMENT OF SOFTWARE ENGINEERING

CLOUD COMPUTING LAB

LAB 01

NAME:

Ayesha Kanwal

REG NO:

FA24B1-SE-077

COURSE CODE:

SE-665

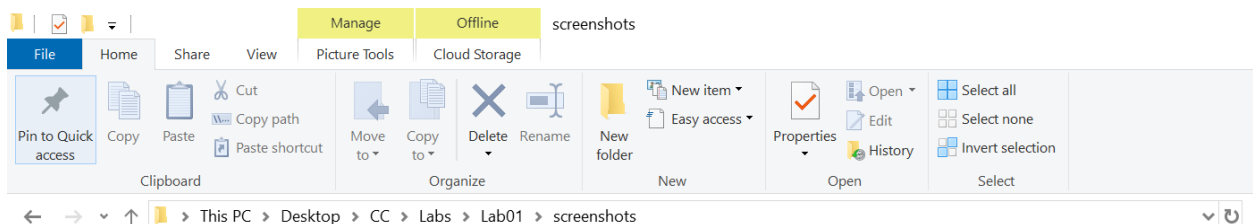
SECTION:

A

GITHUB USER NAME:

ayeshakanwal077

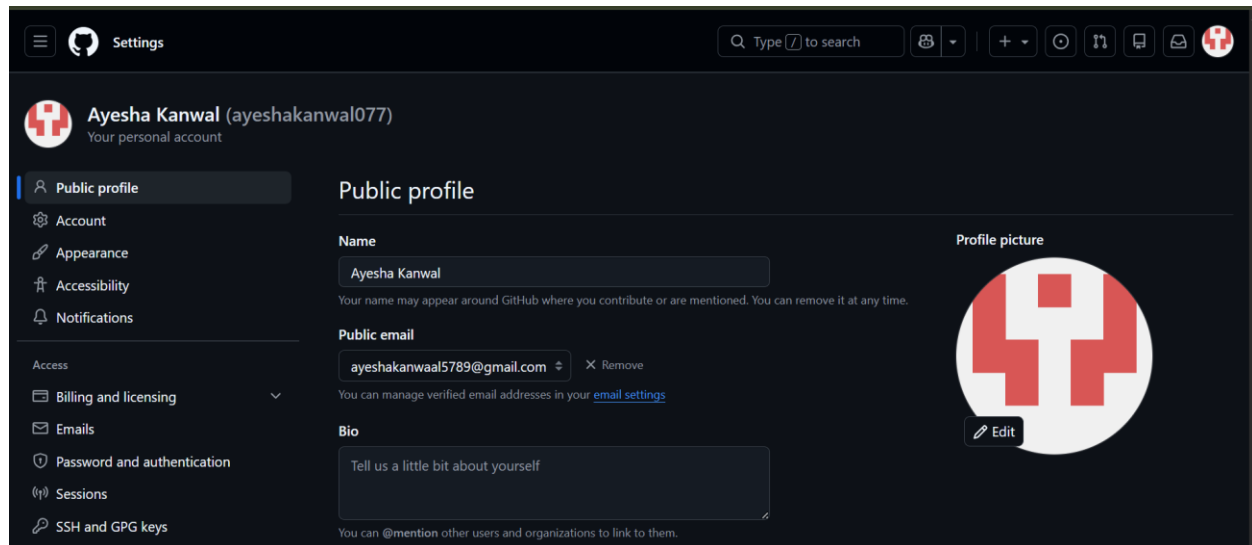
Getting Started



lab01_folder_structure.png

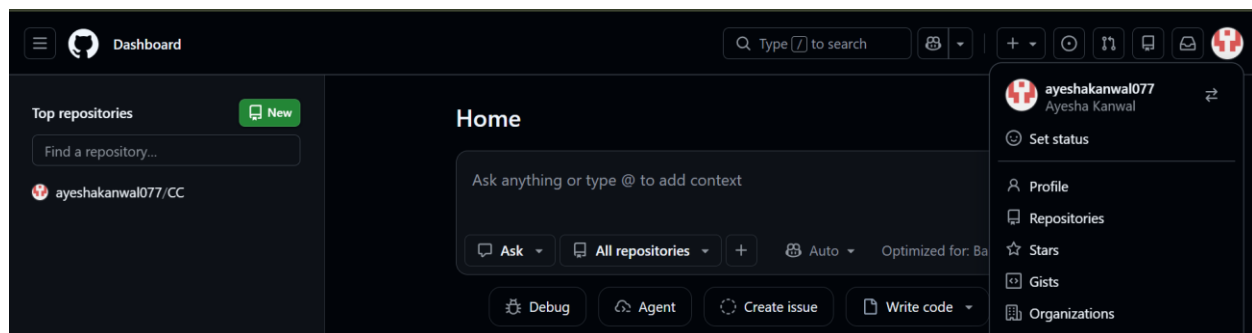
Task 1: GitHub Account Setup

- **Profile Creation:** Logged into GitHub and successfully created a public developer profile under the username ayeshakanwal077.



github_profile.png

- **Email & Security:** Verified the primary account email address and configured basic security settings to ensure repository access.



github_actual_name.png

Task 2: Student Marks Portal Registration

- **Portal Navigation:** Opened the university's official Student Marks Portal login interface in the web browser.

The screenshot shows a web form titled "Complete registration" with a dark circular "ID" icon at the top. Below the title, it says "Enter your registration number, course and section. The enrollment requires approval." The form contains three input fields: "Registration number" with the value "2024-BSE-77", "Course" with the value "SE-665 — Cloud Computing", and "Section" with the value "Section A". Below these fields are two buttons: "Submit enrollment" and "Sign out". A small note below the registration number field states "Allowed: 2024-BSE-01 to 2024-BSE-99."

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-77

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section A

Submit enrollment

Sign out

portal_login_page.png

- **OAuth Authentication:** Connected and authenticated the portal account seamlessly using the registered GitHub credentials.

The screenshot shows a user profile for "Ayesha Kanwal" with a GitHub logo icon and the registration number "2024-BSE-77". The page is titled "Course enrollments" and has a "Sign out" button in the top right corner. A button labeled "+ Enroll in another course" is located on the right side of the profile card.

STUDENT MARKS PORTAL

Course enrollments

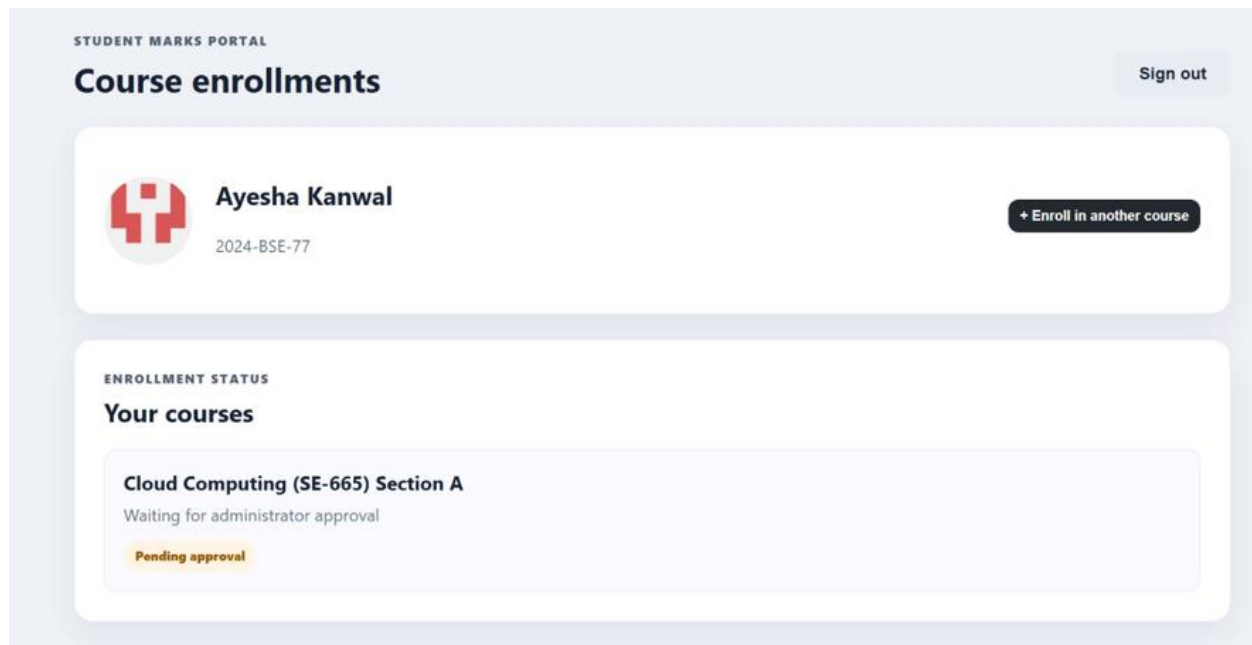
Sign out

 **Ayesha Kanwal**

2024-BSE-77

+ Enroll in another course

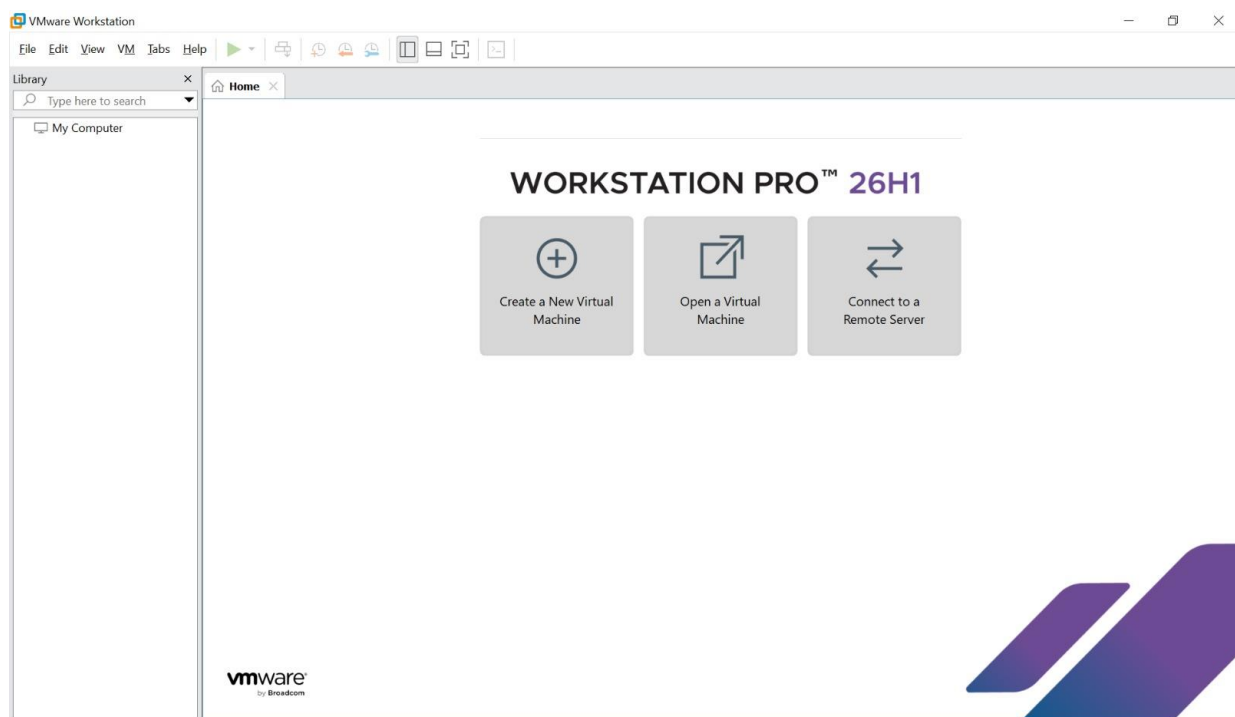
portal_github_profile.png



portal_enrollment_submitted.png

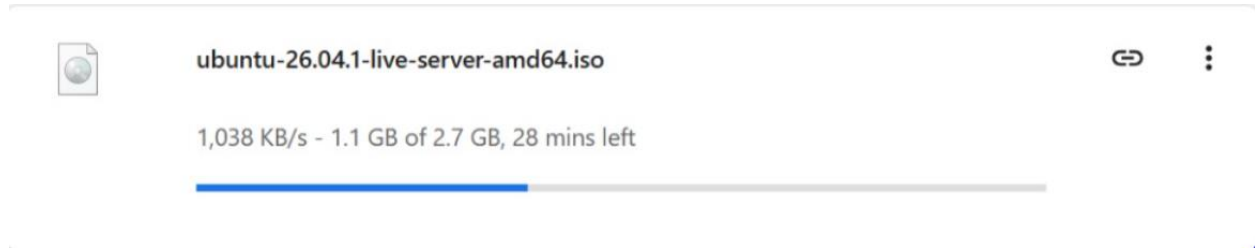
Task 3: Download VMware Workstation Pro and Ubuntu Server

- **VMware Workstation Pro Verification:** Downloaded and installed VMware Workstation Pro on the host Windows machine, opened the application dashboard, and verified .



vmware_workstation.png

- **Ubuntu Server ISO Acquisition:** Navigated to the official Ubuntu download portal and acquired the latest Long Term Support (LTS) Ubuntu Server installation image file.



ubuntu_server_iso.png

- **Storage Availability Check:** Verified the host disk partitions to ensure that a minimum of 20 GB of free storage space is readily available for allocating the virtual machine's hard disk.

Local Disk (C:) - 140 GB



This is how your storage is used and how you can free up space.

available_storage.png

Task 4: Create the Ubuntu Server Virtual Machine

- **New Virtual Machine Wizard Initialization:** Opened VMware Workstation Pro and initiated the virtual machine creation process by selecting the option to create a new virtual machine.



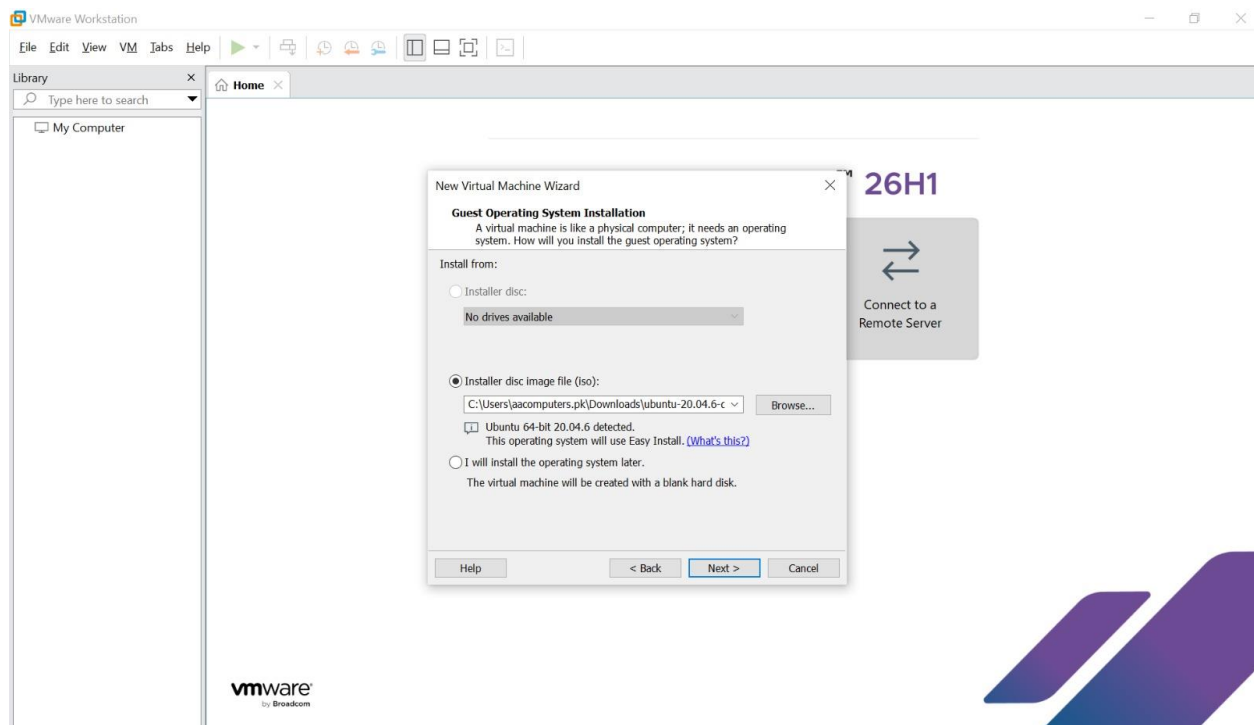
vm_creation_wizard.png

- **Virtual Machine Configuration Type:** Selected the **Typical (recommended)** configuration option inside the wizard to streamline the setup pathway.



vm_typical_configuration.png

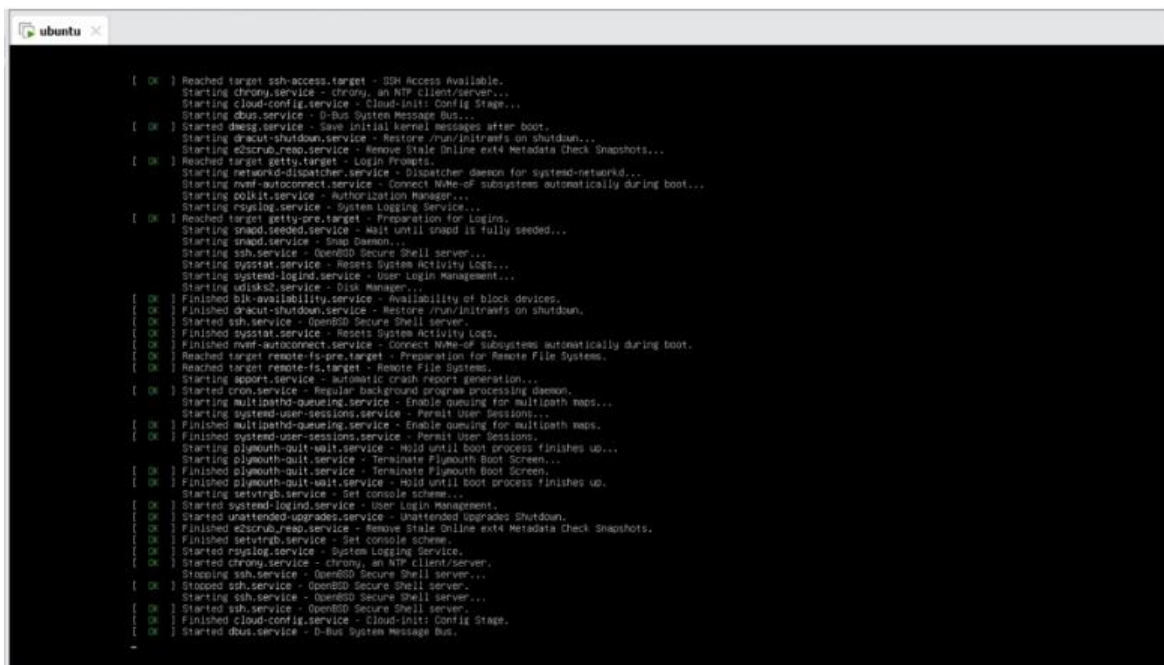
- **Ubuntu ISO Image Selection:** Chose the **Installer disc image file (ISO)** option and browsed to specify the path of the downloaded Ubuntu Server ISO file from Task 3.



vm_ubuntu_iso_selected.png

-
- VMware Workstation
- File Edit View VM Tabs Help
- Library
- Home
- Type here to search
- My Computer
- New Virtual Machine Wizard
- Ready to Create Virtual Machine**
- Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.
- The virtual machine will be created with the following settings:
- | | |
|-------------------|--|
| Name: | ubuntu |
| Location: | C:\Users\aacomputers.pk\OneDrive\Documents\Virtual Ma... |
| Version: | Workstation 25H2 or later |
| Operating System: | Ubuntu 64-bit |
| Hard Disk: | 20 GB, Split |
| Memory: | 2048 MB |
| Network Adapter: | NAT |
| Other Devices: | 2 CPU cores, CD/DVD, USB Controller, Sound Card |
- Customize Hardware...
- ☒ Power on this virtual machine after creation
- < Back Finish Cancel
- 26H1
- Connect to a Remote Server
- vmware
by Broadcom

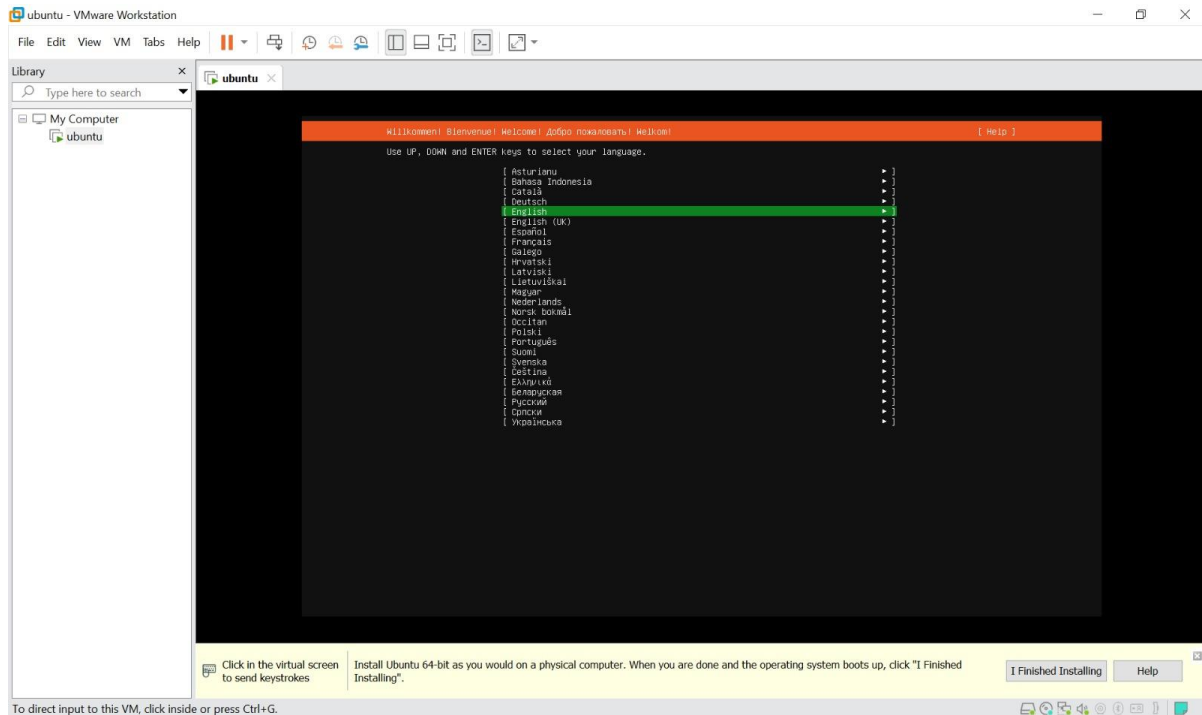
- **Virtual Machine Boot Sequence:** Powered on the newly created virtual machine and waited for the system to boot up into the Ubuntu Server installer screen.



ubuntu installer boot.png

Task 5: Choose the Installation Language

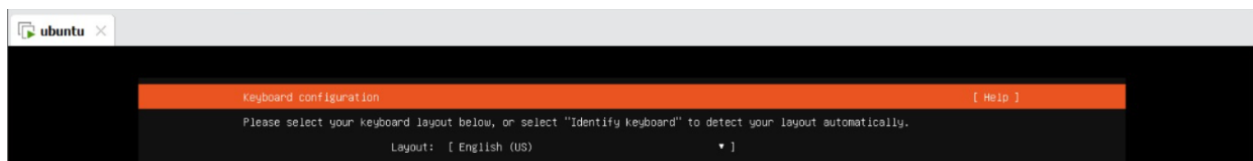
- **Language Selection and Confirmation:** Used the keyboard arrow keys within the boot interface to select the required installation language (English) for the Ubuntu Server setup.



ubuntu_language.png

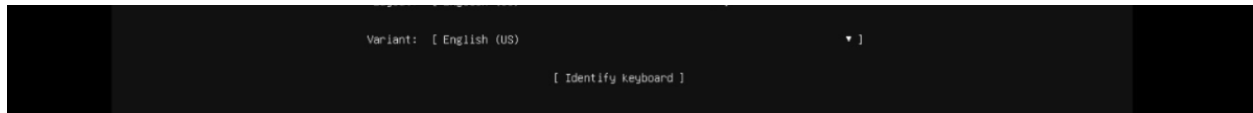
Task 6: Configure the Keyboard Layout

- **Keyboard Layout Selection:** Selected the appropriate keyboard layout matching the physical keyboard configuration from the installer menu.



ubuntu_keyboard_layout.png

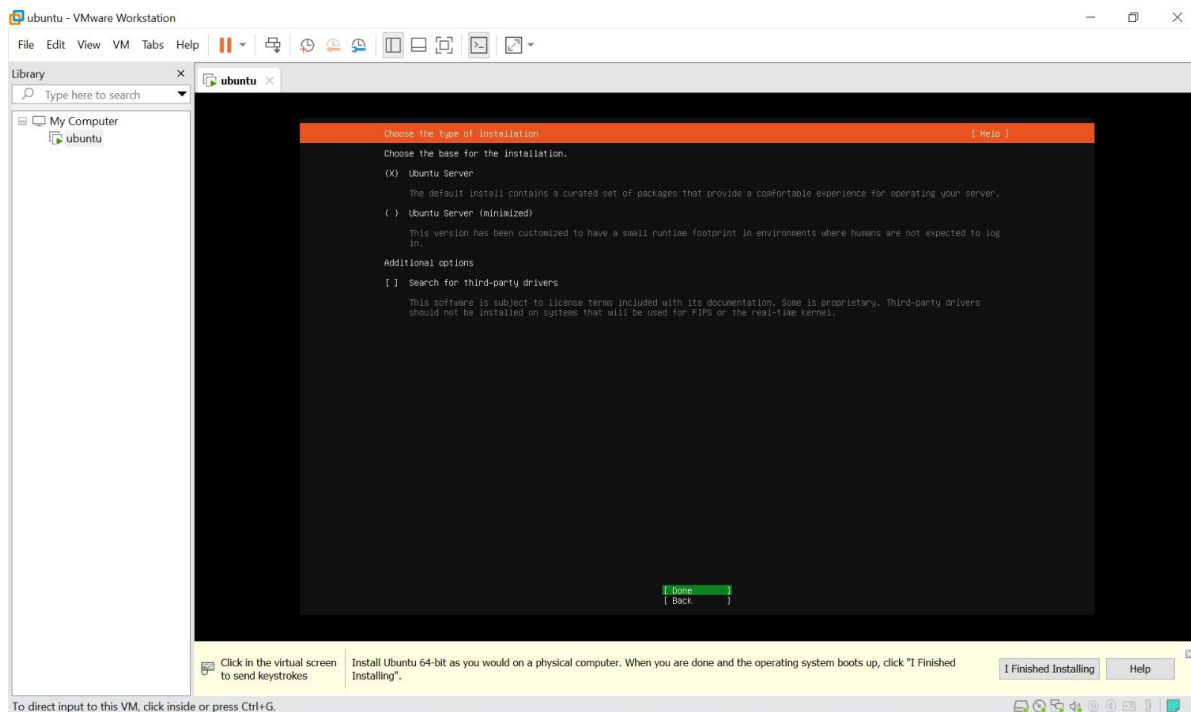
- **Keyboard Variant Configuration:** Chose the specific keyboard variant to ensure correct character mapping and symbol typing during terminal use.



ubuntu_keyboard_variant.png

Task 7: Choose the Ubuntu Server Installation Type

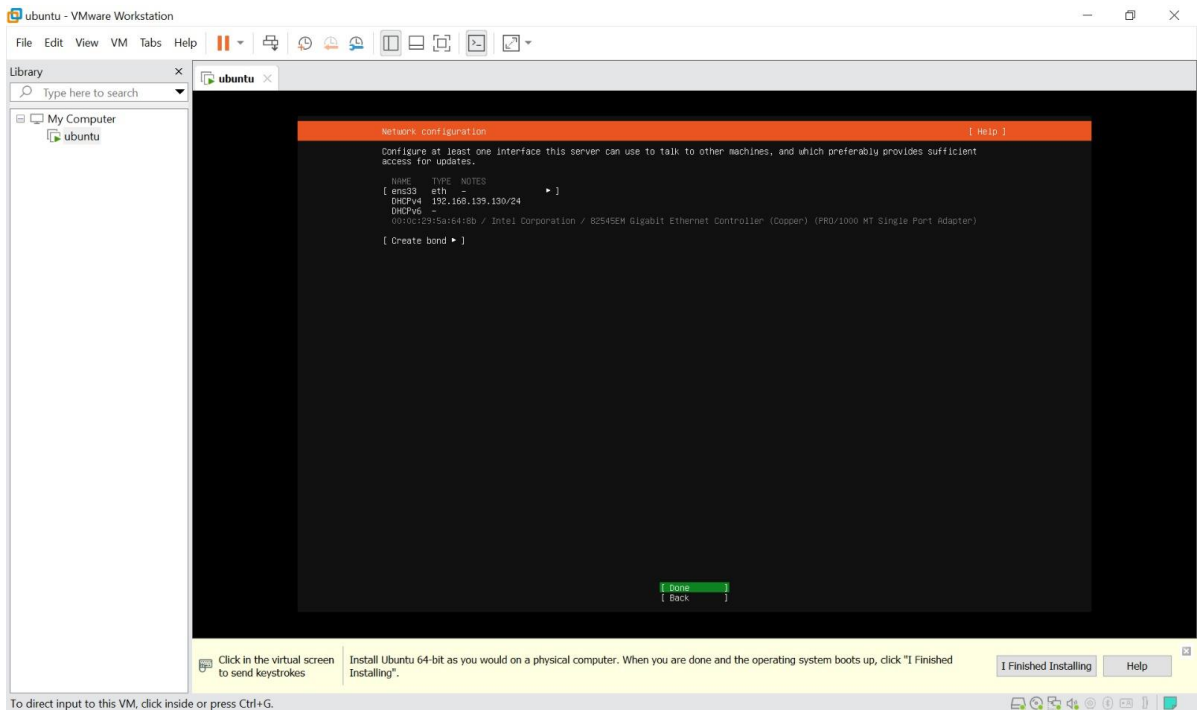
- **Installation Type Selection:** Selected the standard normal Ubuntu Server installation option from the setup menu while avoiding MAAS installation options.



ubuntu_installation_type.png

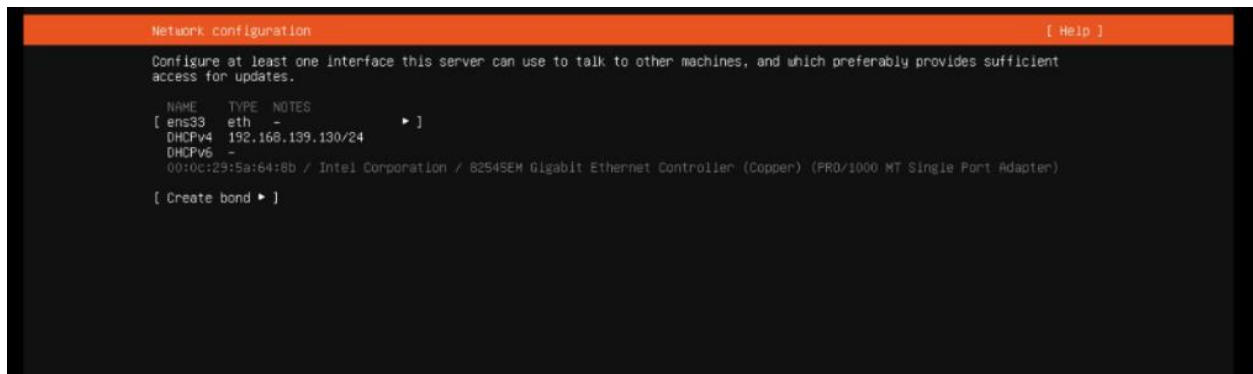
Task 8: Configure the Network

- **Network Interface Detection:** Reviewed the active network interface (ens33) detected by the Ubuntu Server installer during the setup process.



ubuntu_network_interface.png

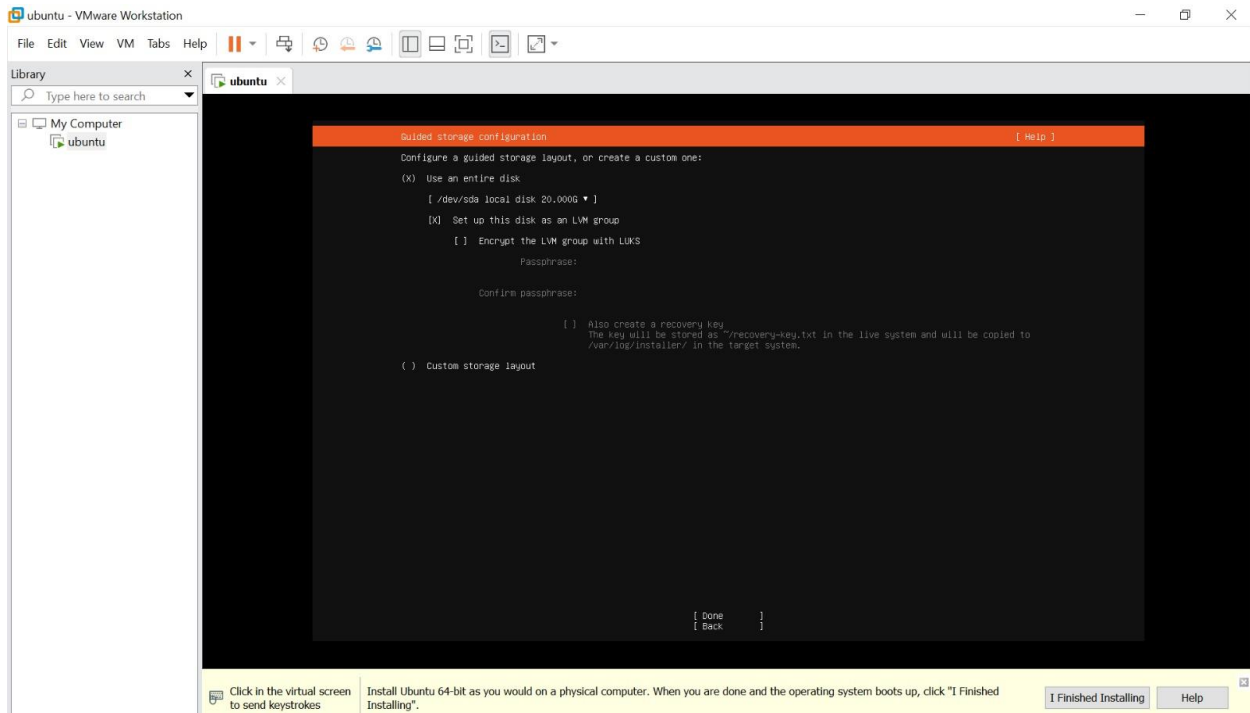
- **DHCP Network Address Assignment:** Confirmed that the network interface successfully received an IP address automatically via DHCPv4 (such as 192.168.121.129/24).



ubuntu_installer_network.png

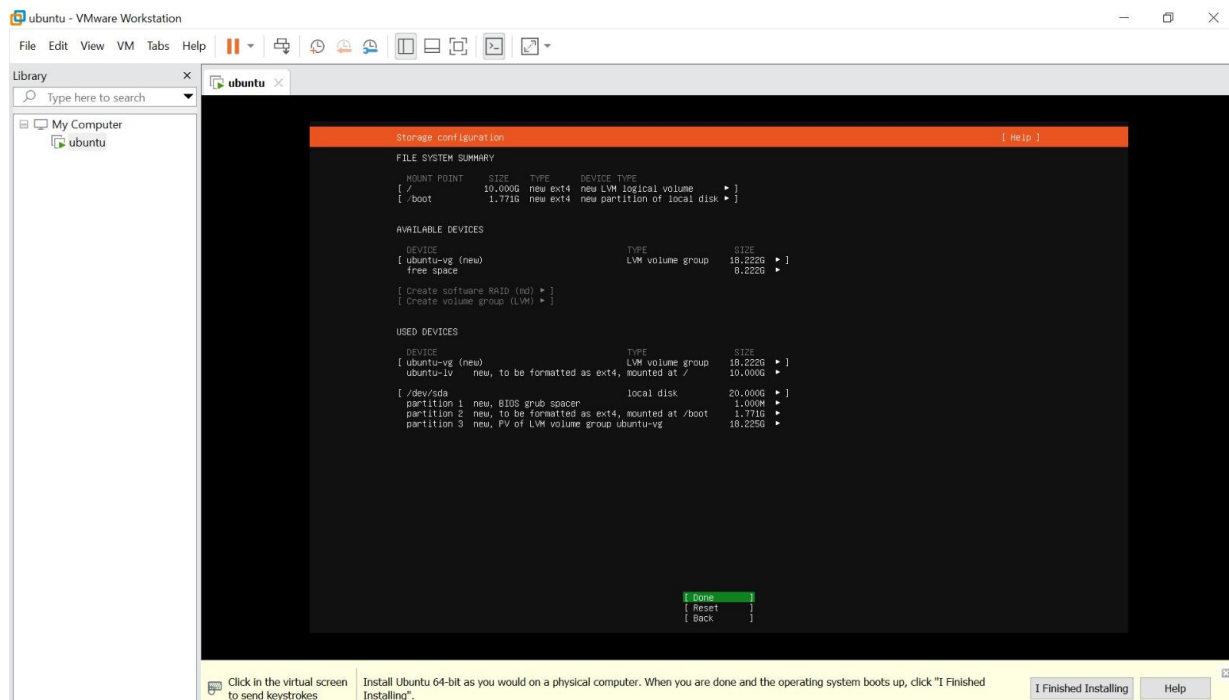
Task 9: Configure Storage

- **Guided Storage Selection:** Selected the guided storage layout option to utilize the dedicated VMware virtual disk created for this lab environment while ensuring no host physical disks are affected.



ubuntu_guided_storage.png

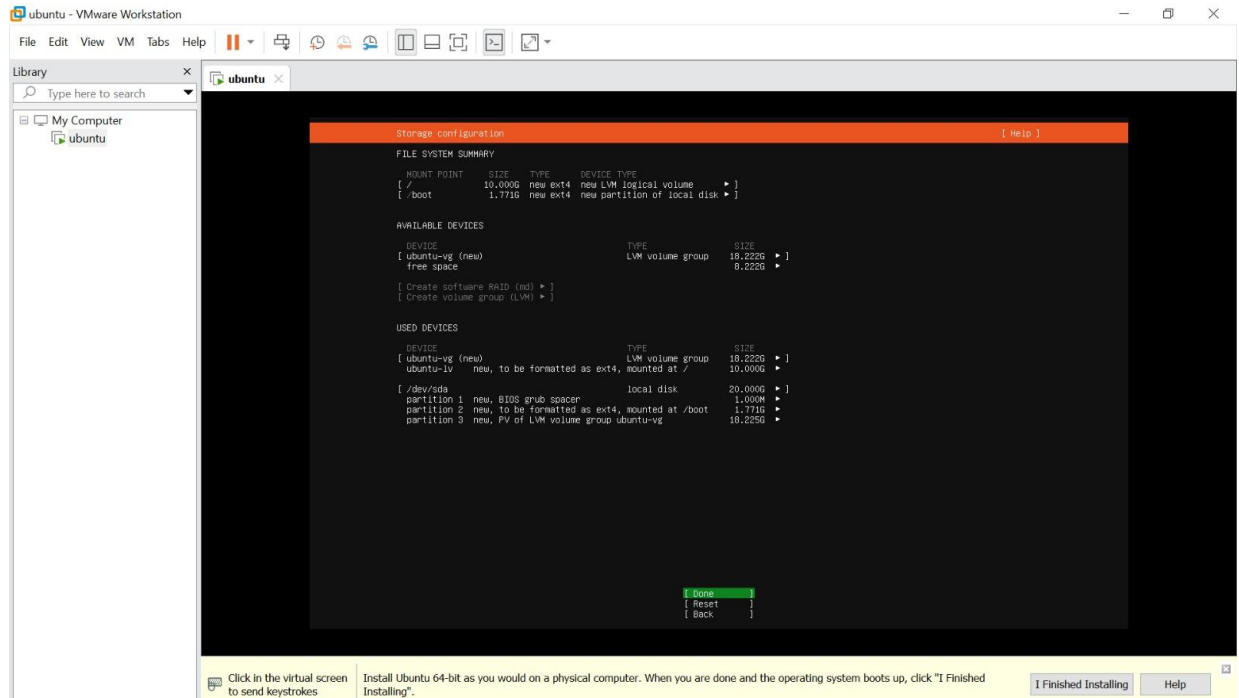
- **Storage Configuration Review:** Reviewed the complete file system summary and partition layout (including the LVM group and mount points) before confirming and proceeding with the installation.



ubuntu_storage_configuration.png

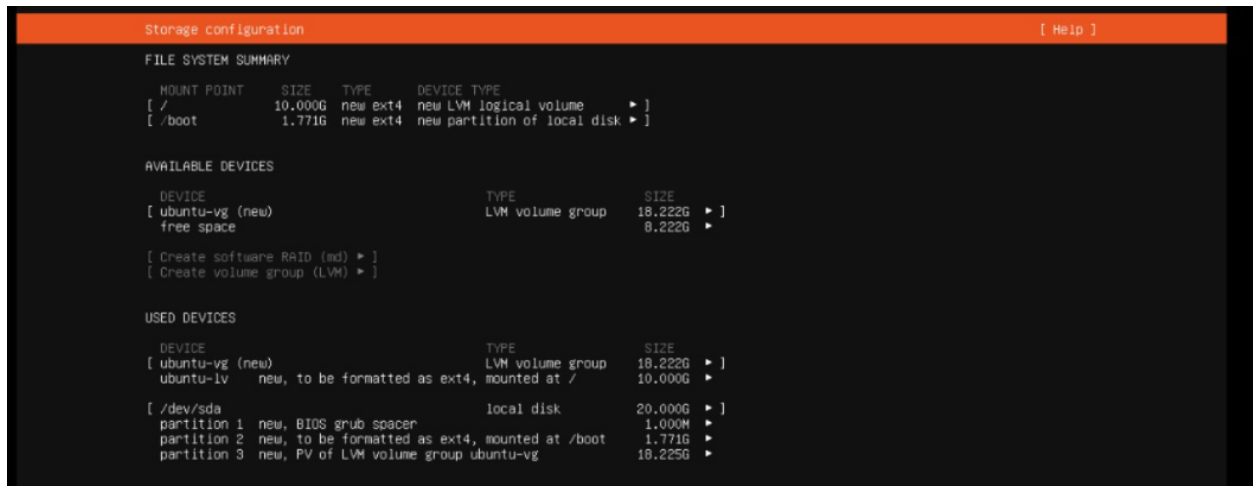
Task 10: Select the Installation Disk

- **Virtual Disk Selection:** Selected the specific virtual disk created for the Ubuntu Server virtual machine during the storage configuration step.



ubuntu_selected_disk.png

- **Disk Size Verification:** Verified the displayed disk name and size to ensure that the correct VMware virtual disk is targeted before proceeding.



ubuntu_disk_details.png

Task 11: Review the Partitions

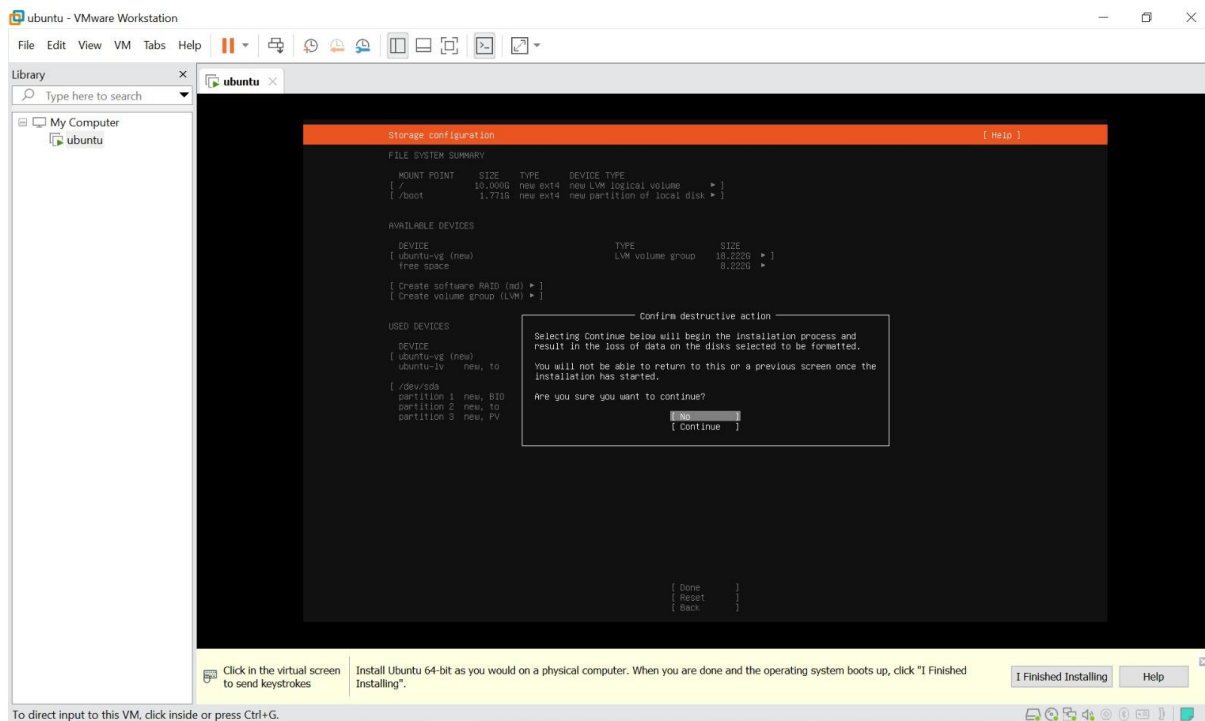
- **Partition Layout Review:** Reviewed the partition layout proposed by the Ubuntu Server installer to ensure the disk structure is properly organized.



ubuntu_partition_layout.png

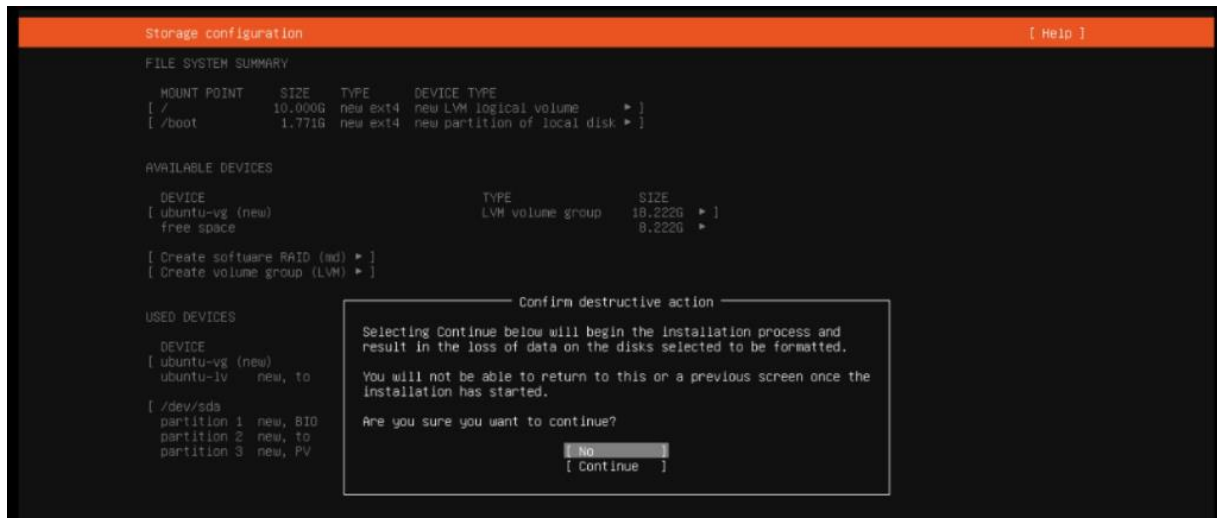
Task 12: Confirm the Storage Changes

- **Destructive Action Warning Review:** Carefully read the destructive-action warning prompt to ensure data safety and verify that only the intended virtual disk is targeted.



ubuntu_storage_warning.png

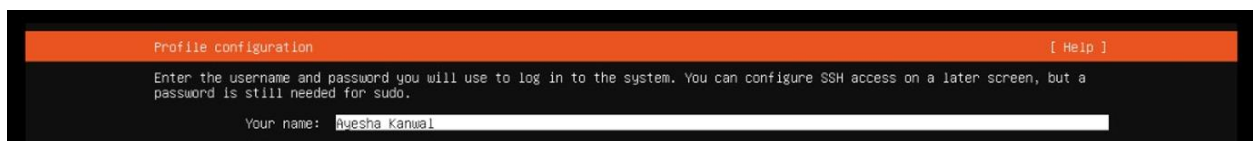
- **Storage Changes Confirmation:** Confirmed and applied the storage configuration changes only after being completely certain that the correct VMware virtual disk was selected.



ubuntu_storage_confirmed.png

Task 13: Configure the Ubuntu Profile

- **User Full Name Entry:** Entered the complete actual name into the "Your name" field of the Ubuntu profile setup configuration.



ubuntu_actual_name.png

- **Server Name Configuration:** Typed ubuntu into the "Your server's name" field to assign the local hostname to the virtual machine.



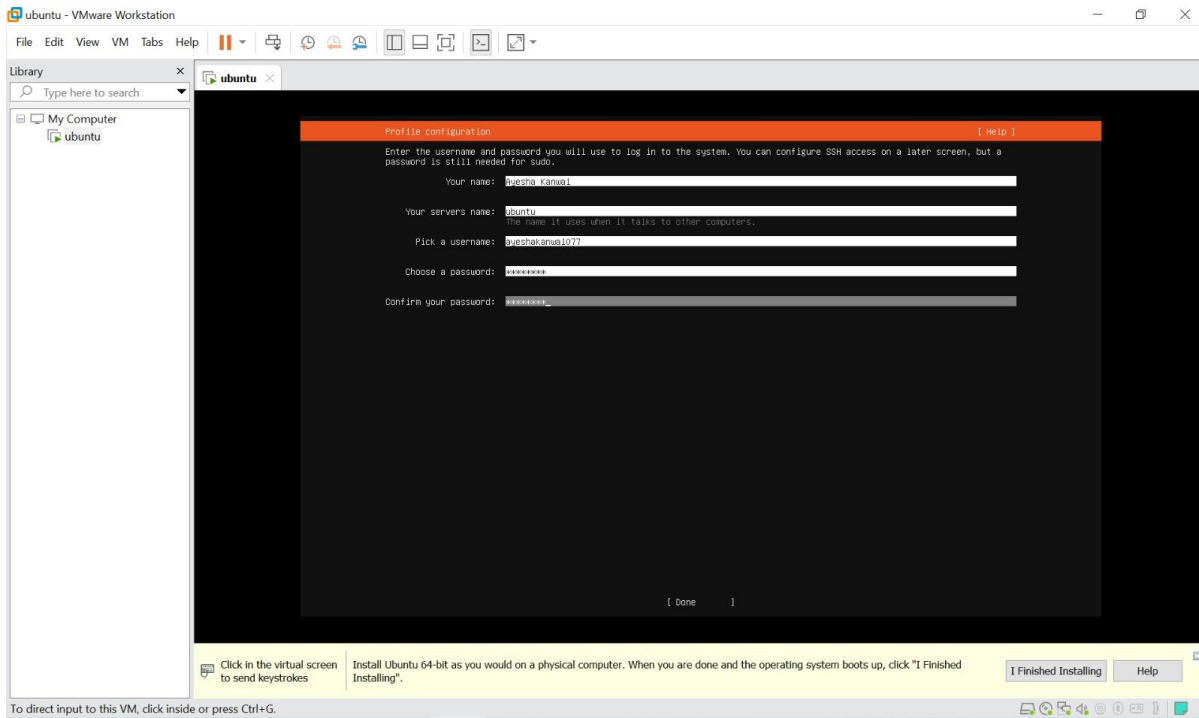
ubuntu_server_name.png

- **System Username Setup:** Entered the exact GitHub username (ayeshakanwal077) into the "Pick a username" field.



ubuntu_username.png

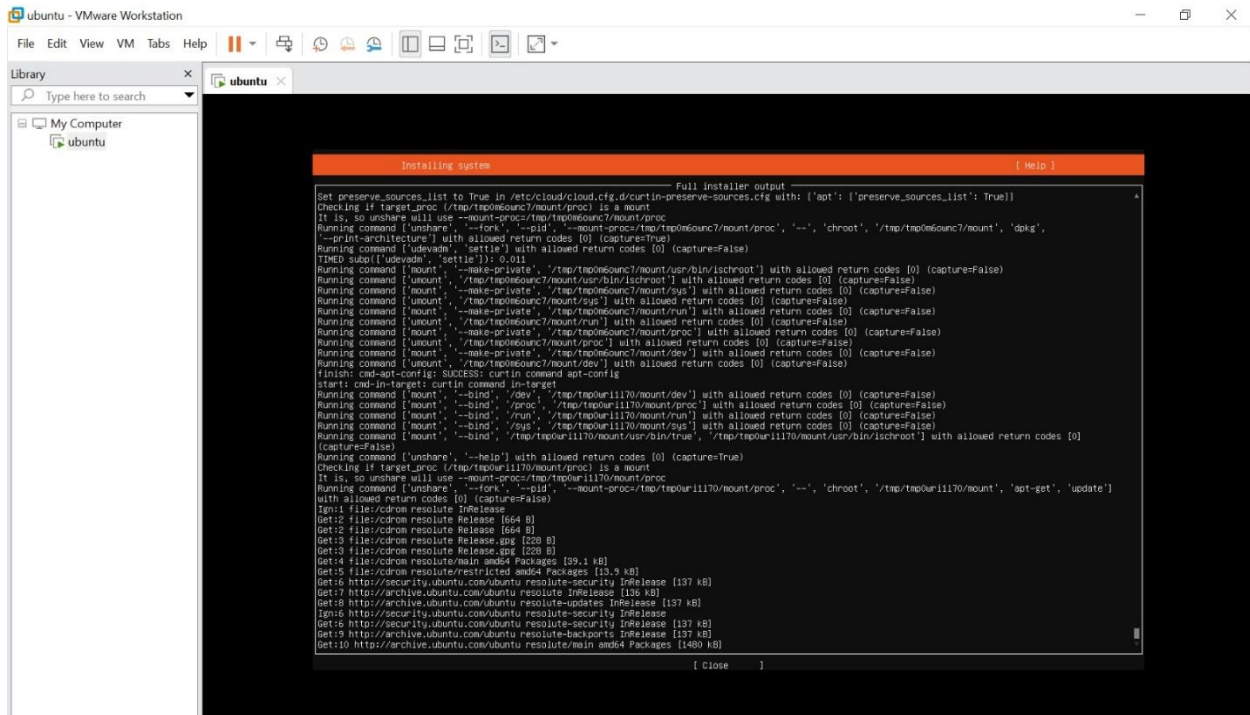
- **Profile Review and Finalization:** Chose and confirmed a secure password for system administration (ensuring credentials were kept hidden from screenshots) and reviewed all profile inputs.



ubuntu_profile_setup.png

Task 14: Complete the Software Installation

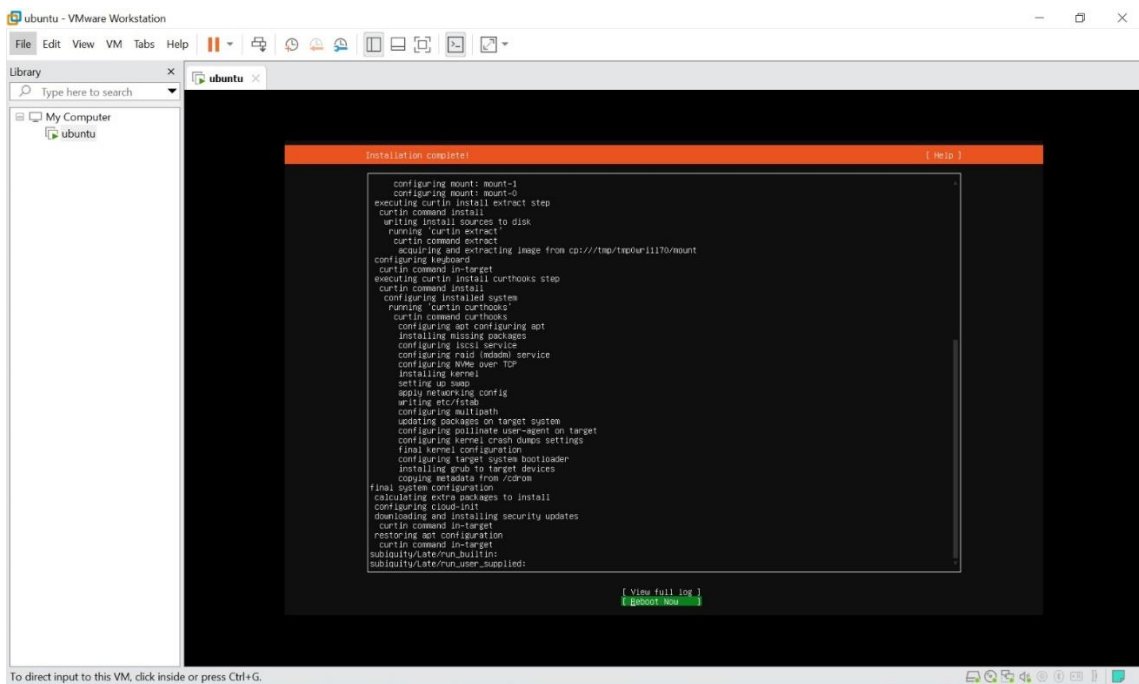
- **Software Installation Progress:** Monitored the system as Ubuntu Server installed all required system packages and core software components, ensuring the virtual machine was not powered off during this critical phase.



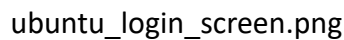
ubuntu_installation_progress.png

Task 15: Reboot and Verify the Ubuntu Identity

- Installation Complete & System Reboot:** Selected the **Reboot Now** option once the software installation completed, disconnected the installer ISO image if prompted by VMware, and allowed the server to restart.

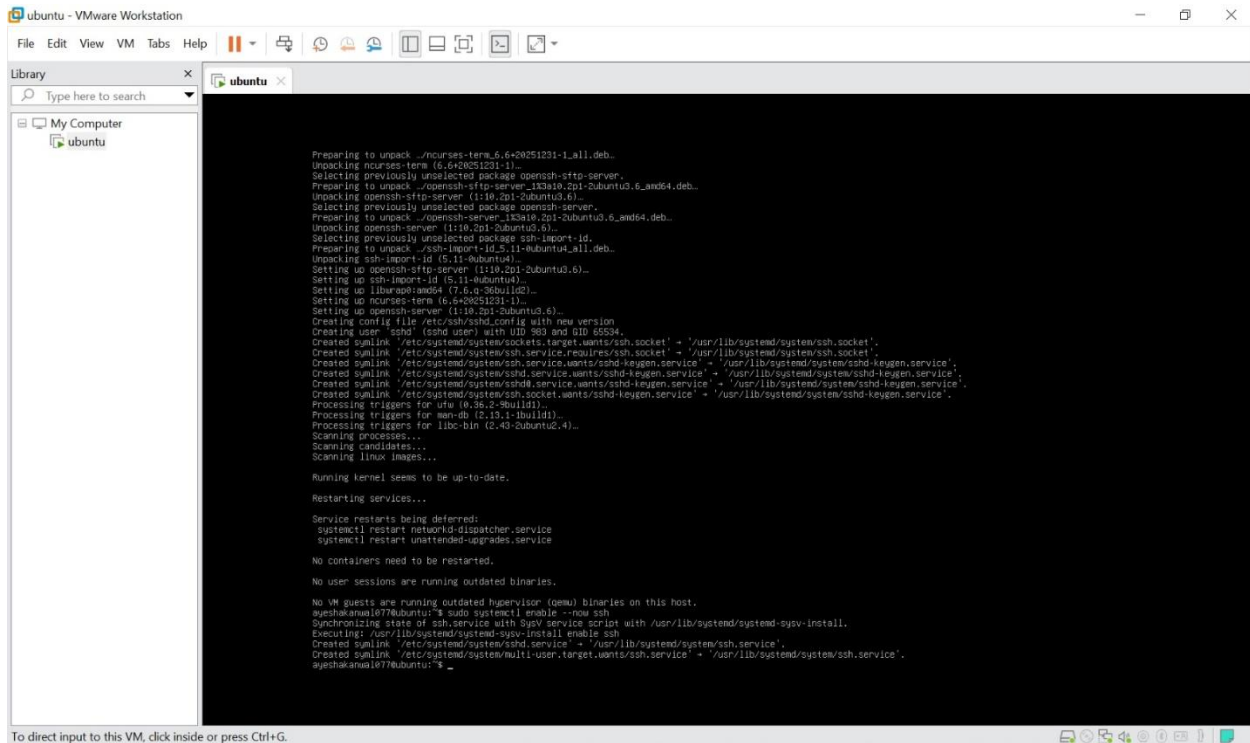


- **Server Login Interface:** Waited for the system boot sequence to finish and captured the initial Ubuntu Server login prompt screen.



- [illegible]

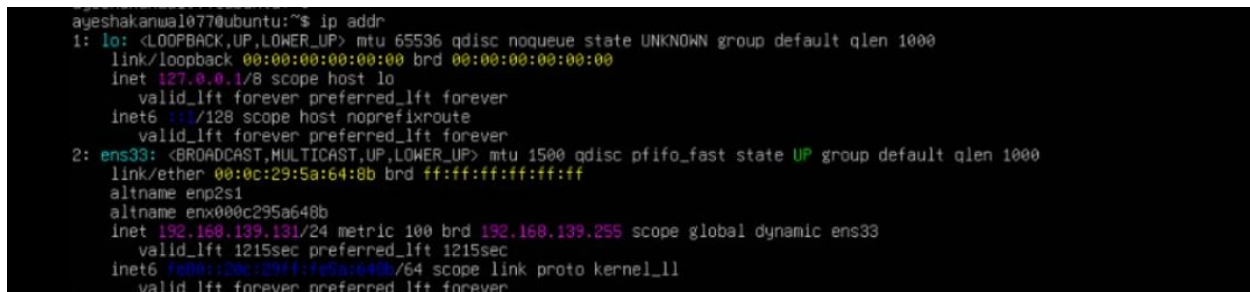
- **Identity Verification via Terminal Prompt:** Verified that the active terminal command line prompt strictly follows the required format .



ubuntu_identity_verified.png

Task 16: Find the Ubuntu Server IP Address

- **IP Address Command Execution:** Executed the networking command `ip addr` inside the Ubuntu Server terminal to inspect all configured network interfaces and address details.



ubuntu_ip_addr_command.png

- **Active IPv4 Address Extraction:** Located the valid non-loopback IPv4 address assigned to the active network interface (avoiding the loopback address 127.0.0.1) while keeping the <github-username>@ubuntu prompt clearly visible.

```

ayeshakanwal077@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:5a:64:8b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c295a648b
    inet 192.168.139.131/24 metric 100 brd 192.168.139.255 scope global dynamic ens33
        valid_lft 1215sec preferred_lft 1215sec
    inet6 fe80::20c:29ff:fe5a:648b/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever

```

ubuntu_ip_address.png

Task 17: Connect to Ubuntu Server from Windows

- **SSH Remote Connection Command:** Opened Command Prompt or PowerShell on the Windows host machine and executed the secure shell command utilizing the correct Ubuntu username and server IP address

```

Command Prompt - ssh ayeshakanwal077@192.168.139.131
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\aacomputers.pk>ssh ayeshakanwal077@192.168.139.131
The authenticity of host '192.168.139.131 (192.168.139.131)' can't be established.
ED25519 key fingerprint is SHA256:XbXPf0iamS6ciHw9vX3f0xE0g/gMJU/NKcAH8Mnz64E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?

```

windows_ssh_command.png

- **Host Fingerprint Acceptance:** Accepted the remote server's cryptographic host key fingerprint by typing yes upon the initial connection prompt.

```

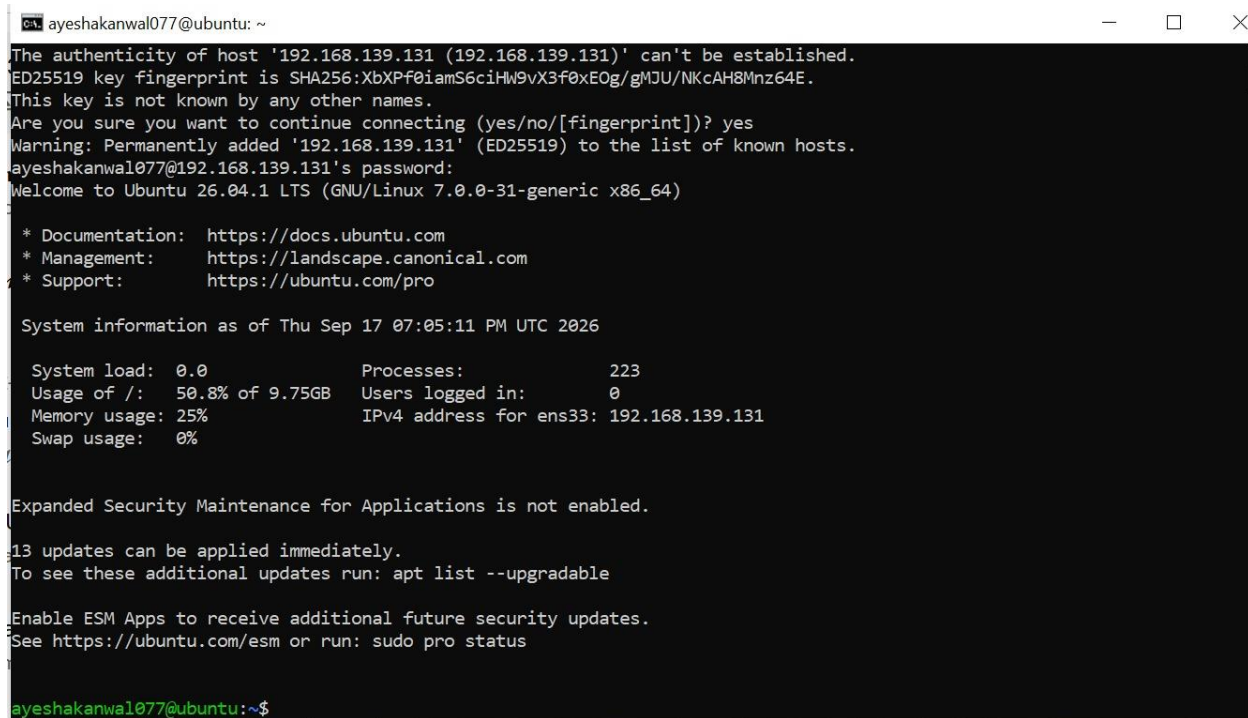
Command Prompt - ssh ayeshakanwal077@192.168.139.131
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\aacomputers.pk>ssh ayeshakanwal077@192.168.139.131
The authenticity of host '192.168.139.131 (192.168.139.131)' can't be established.
ED25519 key fingerprint is SHA256:XbXPf0iamS6ciHw9vX3f0xE0g/gMJU/NKcAH8Mnz64E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.139.131' (ED25519) to the list of known hosts.
ayeshakanwal077@192.168.139.131's password:

```

windows_ssh_fingerprint.png

- **Remote Authentication and Login:** Authenticated securely by entering the Ubuntu account password (with characters hidden for security) to complete the remote login session.



```
ayeshakanwal077@ubuntu: ~  
The authenticity of host '192.168.139.131 (192.168.139.131)' can't be established.  
ED25519 key fingerprint is SHA256:XbXPf0iamS6ciHW9vX3f0xE0g/gMJU/NKcAH8Mnz64E.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.139.131' (ED25519) to the list of known hosts.  
ayeshakanwal077@192.168.139.131's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Thu Sep 17 07:05:11 PM UTC 2026  
  
System load:  0.0          Processes:            223  
Usage of /:   50.8% of 9.75GB Users logged in:          0  
Memory usage: 25%         IPv4 address for ens33: 192.168.139.131  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
13 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
ayeshakanwal077@ubuntu:~$
```

windows_ssh_login.png

- **Remote Shell Identity Verification:** Verified that the active remote terminal prompt correctly displays the expected hostname structure

```
ayeshakanwal077@ubuntu: ~  
The authenticity of host '192.168.139.131 (192.168.139.131)' can't be established.  
ED25519 key fingerprint is SHA256:XbXPf0iamS6ciHW9vX3f0xE0g/gMJU/NKcAH8Mnz64E.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.139.131' (ED25519) to the list of known hosts.  
ayeshakanwal077@192.168.139.131's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Thu Sep 17 07:05:11 PM UTC 2026  
  
System load:  0.0          Processes:            223  
Usage of /:   50.8% of 9.75GB Users logged in:        0  
Memory usage: 25%         IPv4 address for ens33: 192.168.139.131  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
13 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
ayeshakanwal077@ubuntu:~$
```

windows_ssh_identity.png

Task 18: Prepare Lab1 Solution PDF

- **Title Page Generation:** Created the formal lab report document and designed a structured title page containing the lab number, title, complete actual name (Ayesha Kanwal), registration number (FA24B1-SE-077), course code (SE-665), section, and GitHub username (ayeshakanwal077).



DEPARTMENT OF SOFTWARE ENGINEERING

CLOUD COMPUTING LAB

LAB 01

NAME:

Ayesha Kanwal

REG NO:

FA24B1-SE-077

COURSE CODE:

SE-665

SECTION:

A

GITHUB USER NAME:

ayeshakanwal077

solution_title_page.png

- **Document Compilation & Formatting:** Inserted sequential headings for Tasks 1 through 17, systematically placed all corresponding capture files in their correct chronological order with clear descriptive captions underneath each image, and verified overall readability.

